

Damping Improvement of a F-16 Aircraft through Linear Matrix Inequalities

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Abstract: This work presents an application of Linear Matrix Inequalities (LMI) for the robust control of an F-16 aircraft through an algorithm that ensuring the damping factor to the closed-loop system. The results of some tests show that the zero and gain settings are sufficient to ensure robust performance and stability with respect to various operation points. This is one of the main characteristics of LMIs, which are mathematical tools widely applied in control theory, which allow the simultaneous treatment of several performance and robustness requirements. The pole placement is the technique used with the aim to put the system poles in closed-loop in a specific region of the complex plane. Test results using a lateral dynamic model of the F-16 aircraft are presented and discussed.

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1. INTRODUCTION

The dynamic response characteristics of an aircraft are highly non-linear. Generally, flight control systems have been designed using mathematical models of an aircraft, linearized around several operation points, the controller parameters are programmed in accordance with the flight conditions. To the F-16 aircraft control, several techniques have been applied. In Fravolini et al. (2015), a linear strategy control and adaptive control which parameters are calculated by a convex multiobjective optimization, in order to ensure at the same time, the evolution of the error within a minimum invariant set while the linear gain minimized, is performed and applied to the longitudinal dynamic model of the F-16 aircraft. The longitudinal model of a hypersonic flight vehicle was also used for evaluation of the implementation of a robust adaptive controller in Xu et al. (2014), the methodology of this study addresses the issue of controller design and stability analysis in relation to parametric model uncertainties and saturations entrance to the oriented model control. In Holhjem (2012), the adaptive control technique $\mathcal{L}_1$ is applied in closed-loop longitudinal F-16 aircraft model linearized around an operating point. In order to guarantee stability and performance of the resulting gain-scheduled controllers, analytical frameworks of gain scheduling have been developed including the technique of linear-parameter-varying (LPV) control in Biannic et al. (1997), Lu et al. (2005), Crater and Shamma (1996) and Spillman (2000). An application of a conditional integrator based sliding mode control design for robust regulation of minimum-phase nonlinear

systems to the control of the longitudinal flight dynamics of an F-16 aircraft is made by Seshagiri and Promptun (2008). In Liao et al. (2002), a reliable robust tracking controller design method is developed based on the mixed linear quadratic (LQ)/ H_∞ tracking performance index and multiobjective optimization in terms of linear matrix inequalities.

Among the techniques presented for control of an F-16 aircraft, the linear matrix inequality become a possible tool in finding solutions for various optimization problems, control systems and recently identification systems. One of the great advantages of this approach is to allow the simultaneous treatment of various performance and robustness requirements. This is because of the emergence of interior point algorithms for the solution of convex optimization problems, which made it possible to numerically solve the linear matrix inequalities faster and more efficiently.

This paper presents the application of linear matrix inequalities for robust control of an F-16 aircraft. Based on the algorithm presented and developed by Campos et al. (2012), there is the guarantee of the damping factor for the closed-loop system for various operating points by allocating system poles using a predefined controller.

The paper is organized as follows. Section 2 we describe the linear mathematical lateral aircraft model and its general characteristics. This section is extracted mostly from Stevens et al. (2015), with the Simulink model for simulation purposes based on Russell (2003). In section 3, will be presented the mathematical formulation of the

controller structure for the system of F-16 aircraft. Section 4 presents the linear matrix inequalities for pole placement in a particular region of the complex plane, considering several operating points. Results of tests and simulations performed by applying the robust controller to the lateral dynamic model of the F-16 aircraft are presented in section 5. In the last section we present the conclusions of this work.

2. LATERAL DYNAMIC MODEL OF AN F-16 AIRCRAFT

The flat-earth, body-axis 6-Degrees of Freedom (6-DOF) nonlinear control-oriented model for the F-16 fighter aircraft presented in Stevens et al. (2015) and Russell (2003) has been employed in this paper. The nonlinear model is linearized around the operating points (altitude = 4,57 km; total velocity = 549 km/h), and decoupled, to obtain separate longitudinal and lateral-directional linear models. The properties of F-16 aircraft considered in this work are the same in Stevens et al. (2015), and only the lateral-directional, low fidelity state-space model given by (1) is investigated further under the influence of aileron and rudder control inputs.

$$\begin{aligned} \begin{bmatrix} \dot{\phi} \\ \dot{\beta} \\ \dot{P} \\ \dot{R} \\ \dot{\delta}_a \\ \dot{\delta}_r \end{bmatrix} &= A \cdot \begin{bmatrix} \phi \\ \beta \\ P \\ R \\ \delta_a \\ \delta_r \end{bmatrix} + B \cdot \begin{bmatrix} \delta_a \\ \delta_r \end{bmatrix} \\ \begin{bmatrix} \phi \\ \beta \\ P \\ R \end{bmatrix} &= C \cdot \begin{bmatrix} \phi \\ \beta \\ P \\ R \end{bmatrix} + D \cdot \begin{bmatrix} \delta_a \\ \delta_r \end{bmatrix} \end{aligned} \quad (1)$$

Where ϕ , β , P , R , δ_a , δ_r are the aircraft's roll angle (*rad*), angle of side-slip (*rad*), roll rate (*rad/s*), pitch rate (*rad/s*), aileron and rudder deflections (*rad*) respectively. The matrix A, B, C and D can be found using the Simulink program based on Russell (2003). The eigenvalues, the damping ζ , the natural frequency w (*rad/s*) and the overshoot (%) of the lateral dynamic model (1) are shown in table 1.

Table 1. Properties of lateral f-16 aircraft dynamic model in open loop

Eigenvalues	Damping ζ	w (<i>rad/s</i>)	Overshoot (%)
-0.0113	1.0	0.0113	0
-2.12	1.0	2.12	0
-0.32 + 2.74.i	0.116	2.76	69.3
-0.32 - 2.74.i	0.116	2.76	69.3
-20.2	1.0	20.2	0
-20.2	1.0	20.2	0

As we can see, the lateral model has poles at $-0.32 \pm 2.74.i$, which shows that the system has insufficient damping. Our goal will be to improve the system damping, whether it be only in an operating point or not. The theory for this will be presented in the next section.

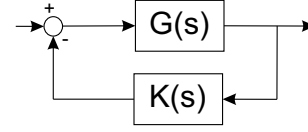


Fig. 1. Structure of the closed loop system

3. SYSTEM CLOSED-LOOP STRUCTURE AND PREDEFINED CONTROLLERS

The theory presented here is based on Campos et al. (2012). The fundamental equations that define the physical behavior of any system linearized about an operating point has the following generic model:

$$\begin{aligned} \dot{x} &= A \cdot x + B \cdot u \\ y &= C \cdot x \end{aligned} \quad (2)$$

where x is the state vector, y is the output vector, and u is the input vector. The structure of the controller to be used to control the F-16 aircraft is pre-defined, which is an important feature, considering the practical application of control systems. This restricted structure is given by the following transfer function:

$$K_{y_k \rightarrow u_l}(s) = \frac{a_{y_k \rightarrow u_l} \cdot s^2 + b_{y_k \rightarrow u_l} \cdot s + c_{y_k \rightarrow u_l}}{s^2 + (p_1 + p_2) \cdot s + p_1 \cdot p_2} \quad (3)$$

where the notation $y_k \rightarrow u_l$ indicates the controller of the output y of the lateral F16 model, with $k = 1, \dots, r$, where r is the number of system outputs, to the input u , with $l = 1, \dots, p$, where p is the number of system inputs. The poles p_1 and p_2 are pre-determined. In this scheme, we work with pre-defined poles and we have to obtain the gain and the zeros, given by the values $a_{y_k \rightarrow u_l}$, $b_{y_k \rightarrow u_l}$ and $c_{y_k \rightarrow u_l}$ of the controller, constrained to feasible values. Our control method comprises applying an output feedback for the F-16 system. The closed-loop system is given in Fig.1.

$K(s)$ is the matrix of transfer functions of the controllers and $G(s)$ is the matrix of transfer functions of the lateral F-16 system. The matrix of $K(s)$ controllers given by (3) can be rewritten in the form of state space as:

$$\begin{aligned} \dot{x}_c &= A_c \cdot x_c + B_c \cdot y \\ u &= C_c \cdot x_c + D_c \cdot y \end{aligned} \quad (4)$$

To define the matrices of the model above, we can use state space realizations, like those described in Campos et al. (2012). Then, matrices A_c and C_c are the following:

$$A_c = \begin{bmatrix} 0 & 1 & \dots & 0 & 0 \\ k & j & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & \dots & k & j \end{bmatrix} \quad (5)$$

$$C_c = \begin{bmatrix} 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & 1 \end{bmatrix} \quad (6)$$

where $k = -p_1 \cdot p_2$ and $j = (p_1 + p_2)$. Note that the matrices A_c and C_c are predefined matrices, since the poles are no problem variables. Applying the controller (4) to the

system described by (2), we have the following description of the system in closed loop:

$$\begin{bmatrix} \dot{x} \\ \dot{x}_c \end{bmatrix} = \begin{bmatrix} A + B.D_c.C & B.C_c \\ B_c.C & A_c \end{bmatrix} \cdot \begin{bmatrix} x \\ x_c \end{bmatrix} \quad (7)$$

Following standard procedure for the design of dynamic controllers using linear matrix inequalities, the problem (7) rewrite as a static feedback problem output:

$$A_m = \begin{bmatrix} A & B.C_c \\ 0 & A_c \end{bmatrix}, B_m = \begin{bmatrix} B & 0 \\ 0 & I \end{bmatrix} \quad (8)$$

$$C_m = [C \ 0]$$

Consequently, the static controller output is:

$$K_c = \begin{bmatrix} D_c \\ B_c \end{bmatrix} \quad (9)$$

With this redesign, the control problem is equivalent to the following structure:

$$\begin{aligned} \dot{x}_m &= A_m \cdot x_m + B_m \cdot u_m \\ y &= C_m \cdot x_m \end{aligned} \quad (10)$$

where $x_m = [x \ x_c]^T$ and the control law $u_m = K_c \cdot y = K_c \cdot C_m \cdot x_m$. Using the state space description and the matrices of the controller A_c e C_c we can evaluate the matrices A_m , B_m e C_m . Therefore, the resulting control problem can be stated as follows: calculate the static gain feedback output, so that the poles of the closed loop system (10) is located in a particular region of the complex plane.

4. POLE PLACEMENT THROUGH LINEAR MATRIX INEQUALITIES

Linear matrix inequalities are mathematical tools that have various applications in control theory, especially in the robust control area. For purposes of pole placement it is important to define regions in a linear matrix inequality.

4.1 Regions of a Linear Matrix Inequality

A region of a linear matrix inequality is any subset of the complex plane that can be defined as, of Chilali et al. (1999):

$$D = \{z \in \mathbb{C} / L + z.R + \bar{z}.R^T < 0\} \quad (11)$$

where L and R are square real matrices with $L^T = L$ and $\bar{z}$ is the complex conjugate of z . Two important features of the regions of a linear matrix inequality are:

- A real matrix is D -stable, that is, has all of its eigenvalues in the linear matrix inequality region D if and only if a real symmetric matrix Q exists such that:

$$\begin{aligned} L \otimes Q + R \otimes (AQ) + R^T \otimes (QA^T) &< 0 \\ Q &> 0 \end{aligned} \quad (12)$$

where $\otimes$ denotes the Kronecker product.

- Intersection regions of linear matrix inequalities are also regions of a linear matrix inequality.

Two regions of a linear matrix inequality interest in control applications for pole placement are as follows:

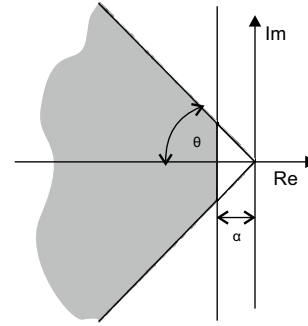


Fig. 2. Complex region plan for closed loop pole placement

- Conical sector with vertex at the origin and interior angle 2θ :

$$L = 0 \text{ and } R = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix}; \quad (13)$$

- Half-plane $\Re(z) < \alpha$:

$$L = 2\alpha \text{ and } R = 1; \quad (14)$$

The region of the formed complex plane (13) and (14) can be seen in Fig. 2.

In this work, the goal is to allocate the system poles of F-16 aircraft in closed loop at the intersection between the sectors of the expressions (13) and (14), ensuring a minimum damping coefficient $\zeta = \cos \theta$, and a transient response with minimum decay rate equal to α for the closed-loop system. Thereby allocating the poles closed loop in this region guarantee adequate performance. To do this, we use the linear matrix inequalities.

4.2 Control through Output Feedback

Applying the control law $u_m = K_c \cdot y$ (with K_c given by (9) to the system (10), we can change the position of the system poles in closed-loop, because:

$$\dot{x}_m = (A_m + B_m \cdot K_c \cdot C_m) \cdot x_m = A_{cl} \cdot x_m \quad (15)$$

In order to put the closed-loop poles in the region described above, we use the results presented in (12). However, the term $(A_{cl} \cdot Q)$ is not linear, since it involves terms of two variables (K_c e Q):

$$A_{cl} \cdot Q = A_m \cdot Q + B_m \cdot K_c \cdot C_m \cdot Q \quad (16)$$

To resolve this issue and transform (16) in a linear matrix inequality problem, it makes a transformation of variables Scavoni et al. (2001):

$$K_c \cdot C_m \cdot Q = N \cdot C_m \quad (17)$$

Substituting (17) into (16) we obtain:

$$A_{cl} \cdot Q = A_m \cdot Q + B_m \cdot N \cdot C_m \quad (18)$$

Note that (18) is a linear equation. Once solved the problem in the transformed variables, the controller gain matrix is retrieved using the following reverse transformation:

$$M \cdot C_m = C_m \cdot Q \quad (19)$$

From the matrix M , the gain matrix K_c is calculated using the following expression:

$$K_c = N \cdot M^{-1} \quad (20)$$

Substituting (18) into (12), we find a set of matrix inequalities that allocate the system poles (10) in the desired region of the complex plan (these inequalities are found in Campos et al. (2012)).

$$\begin{aligned} L \otimes Q + R \otimes (A_m \cdot Q + B_m \cdot N \cdot C_m) + \\ R^T \otimes (Q)_m^T + C_m^T \cdot N^T \cdot B_m^T < 0 \\ Q > 0 \end{aligned} \quad (21)$$

Solving (21) to the resulting sector of the intersection of the two sectors defined by (13) and (14) of the complex plane, it is ensured that the closed loop poles of the aircraft F-16 belong to a desired region of the complex plane. Substituting the values of L and R in the specified regions, is obtained:

- For the conical sector with angle 2θ :

$$\begin{bmatrix} f \cdot A_{cl} \cdot Q + f \cdot Q \cdot A_{cl}^T & g \cdot A_{cl} \cdot Q - g \cdot Q \cdot A_{cl}^T \\ * & f \cdot A_{cl} \cdot Q + f \cdot Q \cdot A_{cl}^T \end{bmatrix} < 0 \quad (22)$$

where $*$ denotes symmetrical term, and:

$$\begin{aligned} A_{cl} \cdot Q &= A_m \cdot Q + B_m \cdot N \cdot C_m \\ Q \cdot A_{cl}^T &= Q \cdot A_m^T + C_m^T \cdot N^T \cdot B_m^T \\ f &= \sin \theta \\ g &= \cos \theta \end{aligned} \quad (23)$$

- Half-plane $Re(z) < -\alpha$:

$$\begin{aligned} 2 \cdot \alpha \cdot Q + A_m \cdot Q + B_m \cdot N \cdot C_m + Q \cdot A_m^T + \\ C_m^T \cdot N^T \cdot B_m^T < 0 \end{aligned} \quad (24)$$

4.3 The Robust Procedure

It was described in the previous section a procedure for pole assignment of the F-16 aircraft system, in a specific region of the complex plane. The system of the aircraft is described by a set of nonlinear equations are linearized around some operating points. However, these operating points only represent the system behavior in a specific condition, and changes in operating points often occur. Thus, it is necessary to ensure that the F-16 will present good performance of the system in case of changes in operating points. To overcome this problem, we will make use of polytopic models.

To set a polytopic model, consider that only the matrix A of the system varies due to changes in operating points of the lateral F-16 aircraft model. Therefore, a polytope Ω is set:

$$\Omega = \{A/A \in R^{n \times n}, A = \sum_{i=1}^m \lambda_i \cdot A_i, \lambda_i \geq 0, \sum_{i=1}^m \lambda_i = 1\} \quad (25)$$

where n is the dimension of the matrices A_i and m the number of operating points. The matrices A_i are called polytope vertices.

To ensure that the poles of any closed loop system associated with the matrix $A \in \Omega$ are in the region of the complex plane defined by (13) and (14), it is necessary to resolve m linear matrix inequalities with the same variables Q and N , in other words:

$$\begin{bmatrix} f \cdot A_{cl,i} \cdot Q + f \cdot Q \cdot A_{cl,i}^T & g \cdot A_{cl,i} \cdot Q - g \cdot Q \cdot A_{cl,i}^T \\ * & f \cdot A_{cl,i} \cdot Q + f \cdot Q \cdot A_{cl,i}^T \end{bmatrix} < 0 \quad (26)$$

for $i=1,2,\dots,m$, with:

$$A_{m,i} = \begin{bmatrix} A_i & B \cdot C_c \\ 0 & A_c \end{bmatrix} \quad (27)$$

$$A_{cl,i} \cdot Q = A_{m,i} \cdot Q + B_m \cdot N \cdot C_m \quad (28)$$

A_i , with $i = 1, 2, \dots, m$, are the matrices in state space that define the mathematical model of the lateral F-16 aircraft, and each of these matrices represent a model not linear linearized around a specific operating point. Solving the system of linear matrix inequalities, it ensures that the system poles in closed-loop are in the region defined by the intersection of the conical sector with semi-plan for all m operating points considered.

5. RESULTS AND DISCUSSION

The method presented in previous section was applied to the dynamic lateral-directional model of F-16 aircraft to the system considering three operation points, the only parameter that changed was the total velocity V to obtain the new linearized system around these operation points, the altitude navigation was kept constant at 4,57 km. The velocities considered are shown in table 2.

Table 2. Operation points

Operation Point	Velocity
1	549 km/h
2	658 km/h
3	768 km/h

For each operation point, new matrices were obtained for the system in the state space, through simulations using the program based on Russell (2003), however, as described in robust method, consider only the variations in matrix A of the system, the matrices B , C and D were considered to be the nominal system (operation point 1).

Initially, according to equation fixed controller (3), the poles of the controller were arbitrarily chosen $p_1 = -5000$ and $p_2 = -5000$, with a value of $\alpha = 0$ and the following performance index (minimum damping coefficient of eigenvalues of the closed loop system):

$$\zeta = \cos \theta = 0.9 \quad (29)$$

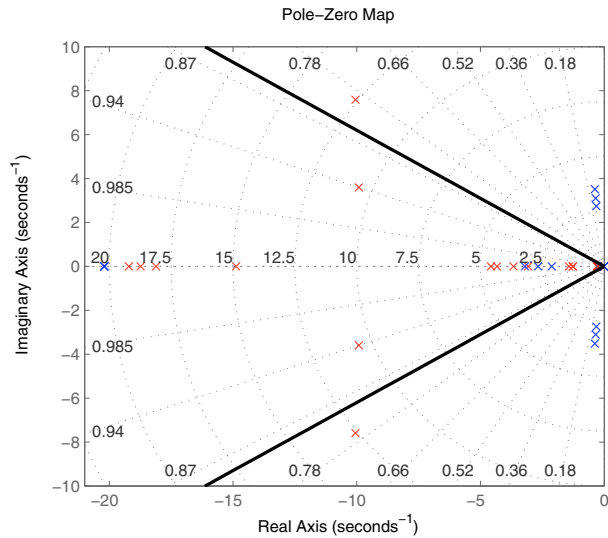
His two zeros were left free, as well as its static gain. The simulation was performed using the software *Matlab*© version R2013b with his toolbox for calculation of linear matrix inequalities. For this case the simulation lasted 3.89 s using a PC with 2.20 GHz, 4GB of RAM, 64-bit. The parameters obtained for the robust controller are shown in table 3. The controller parameters are referenced to the (3).

To evaluate system performance in closed loop, we were plotted all the eigenvalues of the matrices A in closed loop for the three operation points, together with the eigenvalues of the original systems, as can be seen in Fig. 3. In this figure, with two expands, we just show the regions of interest. The black line is an approach to the conic section defined by the angle θ .

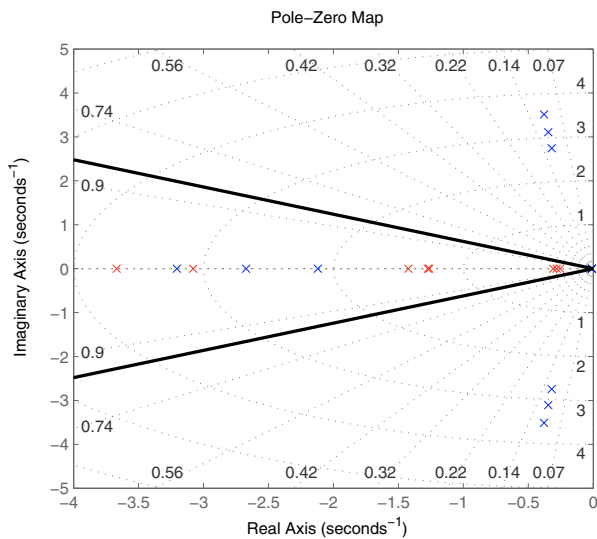
With the simulation of the three operating points, the worst obtained damping factor was $\zeta = 0.8$, with an overshoot of 1.56 %. A pair of poles remained outside the

Table 3. Controller Parameters Found for the Lateral F-16 Control System

y	u	$a_{y_k \rightarrow u_l}$	$b_{y_k \rightarrow u_l}$	$c_{y_k \rightarrow u_l}$
ϕ	δa	0.02283	228.4	5.708×10^5
	δr	-0.07329	-732.9	-1.832×10^6
β	δa	-0.2144	-2143	-5.361×10^6
	δr	0.9115	9114	2.279×10^6
P	δa	0.02347	234.7	5.867×10^5
	δr	-0.06525	-652.4	-1.631×10^6
R	δa	-0.0099	-99.87	-2.475×10^5
	δr	1.268	12680	3.17×10^7



(a) Zoom 1



(b) Zoom 2

Fig. 3. Eigenvalues of F-16 aircraft in open loop (blue) and closed loop with the robust controller (red)

region of linear matrix inequality. This is due to the fact that the process of recovery of the controllers matrices is an approximate process, even with this fact, we verified that the damping factor was higher than the open loop system for all cases. . For this lateral system of F-16 aircraft, other α values were tested, but for a small increase in

its value, the linear matrix inequality became infeasible. Another fact to be noted is that in this case performance specification, the performance obtained by application of the robust controller with this system is that the results obtained were better as the velocity of the F-16 aircraft is increased.

To check system behavior in closed loop with the robust controller, we apply the initial condition $x_0 = [1 \ 1 \ 1 \ 1 \ 1 \ 1]^T$, to the operation nominal point, which showed the worst result among the three operating points, and compared with the open-loop system. Then, we obtain the graph shown in Fig. 4. As can be seen in this image, the system oscillation and settling time improved significantly compared to the system without the robust controller.

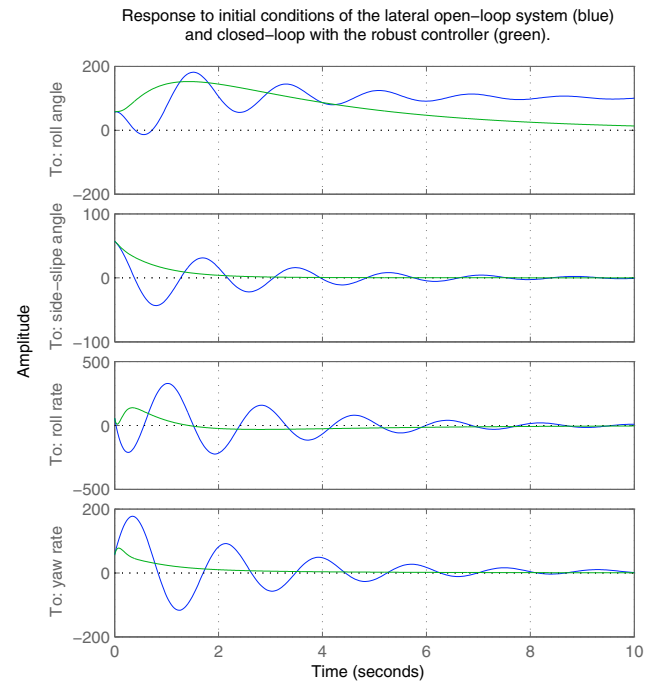


Fig. 4. Response to initial conditions of the lateral open-loop system (blue) and closed loop with the robust controller (green)

To verify the behavior of the robust procedure for several values of the damping factor ζ , the poles of the controllers and the factor α were kept constant and a change in damping value $\zeta = \cos \theta$ was applied, from its worst value for each point of operation to $\zeta = 0.95$. The results were plotted on a graph $\zeta \times \zeta$, which can be observed in Fig. 5.

In the graph obtained, we observe that the operation point 3 first reaches the damping, in approximately $\zeta = 0.5$, while the others, in $\zeta = 0.6$ and 0.8 respectively. As we can see, for all ζ values up to 0.8 , all damping factors found were above the reference, which means that all eigenvalues of the system were within the specified region of the LMI.

Another performance evaluation was performed, except that in this case we keep the α value of the damping factor ζ constant, in which case it is considered the worst case, and we vary the value of the poles p_1 and p_2 , from -30 to -5000 . The graph obtained with this variation can be seen in Fig. 6.

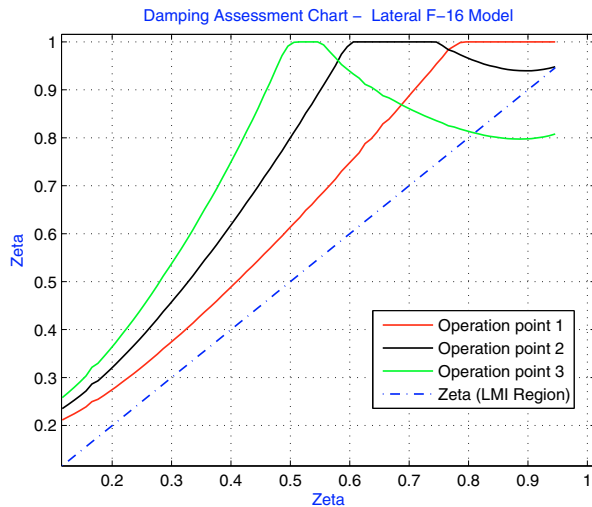


Fig. 5. Damping Assessment Chart - Lateral Model.

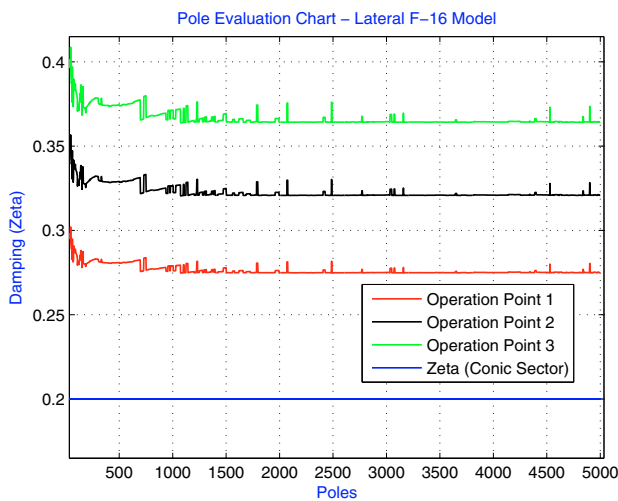


Fig. 6. Pole Assessment Chart - Lateral Model.

As can be seen, improved damping values vary little with the positioning of the controller poles.

6. CONCLUSION

In this paper we present a methodology for pole placement of a linearized system around various operating points at regions of the complex plane, defined by the intersection of two regions of a linear matrix inequalities. The controller used here has a fixed structure, in which initially define the poles and control through linear matrix inequalities, obtain their gain values and their zeros.

It was applied to pole placement through the linear matrix inequalities to lateral F-16 aircraft system. For specified performance conditions, there was an improvement in all the properties considered in this work to the closed-loop system.

As can be seen in the responses to the initial conditions for the system, there is an evident improvement in the response rate and damping output. In addition, the applied controller is robust, which gives it advantages over any

other controller, as it considers various system operating points to be controlled. One of the main advantages of the formulation presented in this work is that it generates controllers with a pre-specified structure, which can be applied to control the F-16 aircraft. This makes it easy to practical implementation to test the controllers obtained on stabilization and increased performance.

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